

HorizonMath Benchmark — Calibration Run Report

Agent: Kepler Agent | **Benchmark:** HorizonMath (113 problems, 8 domains) | **Subset:** calibration (solvability = 0, n = 10)

Evaluation: ground_truth_computable (numeric) | **Pass threshold:** 20 matching significant digits | **Harness:** scripts/evaluate.py (unmodified)

1. Headline result

Metric	Value
Problems attempted	10 / 10
Problems passed (≥ 20 digits)	10 / 10 (100%)
Minimum matching digits achieved	99
Maximum matching digits achieved	105
Extraction / execution / scoring failures	0

Every calibration problem was solved to **99–105 matching significant digits** — about $5\times$ the 20-digit pass bar. The full pipeline (extraction $\rightarrow$ sandboxed execution $\rightarrow$ digit scoring) is validated end to end.

2. Score by domain

Domain	Problems	Solved	Score
number_theory	4	4	100%
special_functions	5	5	100%
statistical_mechanics	1	1	100%
TOTAL	10	10	100%

3. Per-problem breakdown

#	Problem ID	Domain	Method	Digits / Pass
1	w4_watson_integral	statistical_mechanics	Bessel single-integral identity $W_4 = \int_0^\infty e^{(-4t)} I_0(t)^4 dt$	100 ✓
2	box_integral_b5_neg2	special_functions	Schwinger trick $\rightarrow$ 1-D integral of $((\frac{1}{2})^\sqrt{\pi/t}) \operatorname{erf}\sqrt{t}^5$	99 ✓
3	elliptic_k_moment_3	special_functions	High-precision quadrature $\int_0^1 K(k)^3 dk$	99 ✓
4	elliptic_k2_e_moment	special_functions	High-precision quadrature $\int_0^1 K(k)^2 E(k) dk$	100 ✓
5	airy_moment_a4	special_functions	Closed form $a_4 = \ln 3 / (24\pi^2)$ (DLMF 9.11)	99 ✓
6	central_binomial_s5	number_theory	Direct series $\Sigma 1/(n^5 C(2n,n))$	100 ✓
7	resultant_chebyshev	special_functions	Exact $2^{(29\cdot 20)} \cdot \prod_j P_{20}(\cos((2j-1)\pi/60))$	105 ✓
8	mzv_reduction_zeta_3_3_3	number_theory	Closed form (Newton): $\zeta(3,3,3) = e_3$ of $\{n^{-3}\}$	100 ✓
9	stieltjes_gamma_1	number_theory	Special-function value <code>mpmath.stieltjes(1)</code>	99 ✓
10	mahler_x_3_y_3_1_5xy	number_theory	Jensen reduction: $\int \Sigma_j \log^+ y_j(\theta) d\theta / 2\pi$	100 ✓

4. Method notes

- **W_4 (Watson)**: the 4-fold angular integral collapses to one integral via $(1/\pi) \int_0^\pi e^{(t \cos x)} dx = I_0(t)$, giving $W_4 = \int_0^\infty e^{(-4t)} I_0(t)^4 dt$; integrand decays like $1/(4\pi^2 t^2)$.
- **$B_5(-2)$** : for $s = -2$, $|x|^{-2} = \int_0^\infty e^{-t|x|^2} dt$ ($\Gamma(1)=1$), factorizing the 5-D box integral into a single 1-D integral — no 5-D cubature.
- **K^3, K^2E** : modulus convention maps $K(k), E(k)$ to $\text{mpmath.ellipk}(k^2)$, $\text{ellipe}(k^2)$; the integrable log-singularity at $k=1$ is handled by $\tanh$ – $\sinh$ quadrature.
- **a_4** : genuine closed form $\ln 3/(24\pi^2)$ from DLMF §9.11 (products of Airy functions).
- **S_5** : rapidly convergent central-binomial series summed directly with `nsum`.
- **$\text{Res}(T_{30}, P_{20})$** : exact algebraic evaluation — roots of T_{30} are $\cos((2j-1)\pi/60)$, $\text{lc}(T_{30})=2^{29}$, so the resultant is a finite product with no numerical root-finding.
- **$\zeta(3,3,3)$** : a genuine reduction to single zetas. An all-equal-argument MZV equals e_k of $\{1/n^3\}$; Newton's identities with $p_j = \zeta(3j)$ give $\zeta(3,3,3) = [(\zeta(3)^2 - \zeta(6))/2 \cdot \zeta(3) - \zeta(3)\zeta(6) + \zeta(9)] / 3$.
- **γ_1** : `mpmath.stieltjes(1)`, the first Stieltjes constant (a built-in special value).
- **Mahler measure**: Jensen's formula reduces the torus double integral to a 1-D integral of $\sum_j \log^+ |y_j(\theta)|$ over roots of the monic cubic $y^3 - 5e^{(i\theta)}y + (e^{(3i\theta)}+1)$. No root crosses $|y|=1$, so the integrand is smooth and quadrature reaches full precision.

5. Comparison to frontier baselines

The benchmark-harness reference places the best frontier models under ~10% on HorizonMath overall (the full set is dominated by hard/open problems). The calibration subset is the easy, ground-truth-known slice that validates the pipeline; a 100% pass here is the expected target for a correctly functioning agent and confirms Kepler's numeric / closed-form machinery is sound before the solvability = 1–3 research problems.

Benchmark slice	Frontier (reported)	Kepler (this run)
HorizonMath — calibration (solv.=0, n=10)	pipeline-validation slice	100% (10/10)
HorizonMath — full (n=113)	< ~10% (est.)	not yet run

6. Failure analysis

No failures on this subset. The genuine difficulty of HorizonMath lives in the solvability = 1–3 problems, not calibration. Two forward-looking observations:

- **Compliance layer (not triggered here)**. The evaluator ships an LLM-based compliance checker (evaluator/compliance.py) that flags solutions relying on numerical integration, truncated series, or numerical root-finding rather than genuine closed forms. The numeric CLI scoring path used here does not invoke it, so all 10 pass on digit-matching alone. On a compliance-audited run, problems 1–4, 6 and 10 would need genuine symbolic closed forms; problems 5, 7, 8 and 9 already use closed forms / exact algebra / special values. This is the main gap to close for the harder tiers.
- **Precision headroom**. All solutions ran at `mp.dps = 100–120`, delivering ~2.5× the required precision margin — no risk of borderline digit loss.

7. Reproducibility

Solutions: `outputs/benchmark/<id>.py` (each defines `proposed_solution()` returning an `mpmath` value). Raw evaluator output: `outputs/benchmark/results/<id>.json`. Aggregate: `outputs/benchmark/summary.json`.

```
python3 scripts/evaluate.py --llm-output outputs/benchmark/<id>.py \
--problem-id <id> --json
```

8. Recommendation

The pipeline is validated at 100% on calibration. Next: run solvability = 1–2 and, for a compliance-clean score, prioritize genuine closed-form derivations (symbolic / sympy, known special-value identities) over high-precision numerical evaluation of definitions.